

# STREAMLIT APP IMPLEMENTATION & QA.

## AUDITING APP.PY AGAINST THE MODULE 9 DEPLOYMENT PLAN

8 pages, 724 lines of Streamlit — checked page by page against the three build risks Module 9 flagged in advance, plus anything new found in the code.



AI LAB 99

| 8-PAGE STATUS AT A GLANCE |         |  |                     |
|---------------------------|---------|--|---------------------|
| Page                      | Status  |  | Page                |
| Home                      | OK      |  | Customer Prediction |
| Dashboard                 | BROKEN  |  | Visualizations      |
| Customer Segments         | OK      |  | Marketing Strategy  |
| Customer Search           | PARTIAL |  | Download            |

✓ HOME — OK

Customer/feature/segment counts are raw counts, not scaled values — safe as-is.

⚠ DASHBOARD — CONFIRMED BUG

if"\${average\_income:;, .0f}" is applied directly to standardized Income (range ≈ −3 to +3). "Average Income" and "Total Spend" show meaningless \$ figures. The Income Range Filter slider truncates the same z-scores to an int range like −1 to 3.

✓ CUSTOMER SEGMENTS — OK

Counts & percentages come from the categorical segment column — unaffected by scaling.

CUSTOMER SEARCH — MINOR ISSUE

Placeholder text suggests searching by "Customer ID" — that column was dropped in Module 5 feature selection and no longer exists in the dataset.

⚠ CUSTOMER PREDICTION — STUB

No model loaded, no input form, no prediction ever runs. Just a status message + warning. The "real-time predictor" in the nav menu predicts nothing.

VISUALIZATIONS — PARTIAL

Charts render correctly but "Income Tier Distribution" and "Total Product Category Revenue" are drawn from un-inverse-transformed z-scores — shapes are directionally fine, axis values and the word "Revenue" overstate what's shown.

⚠ MARKETING STRATEGY — STATIC, CONFIRMED

4 hard-coded cards (Premium, At-Risk, Budget, Digital-First) are **not wired to any data**. Only 2 real clusters exist. Worse: the single second cluster's two historical names get **split into two separate cards** with different strategies.

✓ DOWNLOAD — OK

Full and filtered CSV export both work via st.download\_button.

CODE EVIDENCE

```
st.metric("AVERAGE INCOME",
         f"${'}average_income:,.0f{'}"
         if average_income > 0 else "N/A")
```

Direct \$ formatting on a StandardScaler column, no `inverse_transform()` call anywhere in app.py.

NAMING DRIFT, NOW IN THE UI

Cluster 1 has carried 3 names across modules (Digital-First → Budget → Digital-First) — and the app now shows **both** "Budget & Value Seekers" and "Digital-First Shoppers" as if they were two separate, real customer segments.

WHAT WORKED

3 of 8 pages (Home, Customer Segments, Download) are fully correct because they only ever touch counts or categorical labels — never the scaled numeric columns.

MODULE 9 RISKS — OUTCOME

2 of 3 Materialized

Hit

Z-SCORE \$ DISPLAY

Hit

EMPTY STRATEGY PAGE

Avoided

BAD PREDICTIONS

PRIORITY FIX LIST

1

Inverse-transform

Income/Total\_spending/Recency once at load time using standard\_scaler.pkl

2

Cut Marketing Strategy

to the 2 real segments, one name each

3

Decide on Prediction:

build it for real or remove it from nav

4

Fix Search placeholder:

drop the "Customer ID" hint

5

Add a units disclaimer

to the Download page